

Predicting Home Value based on different demographics within the United States

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Abstract

In this project I am predicting home value based on demographics using R from a dataset I retrieved from IPUMS USA. I will be testing different predictive models to see which returns the lowest MSE. I will be testing Lasso, Logistic Regression, OLS with polynomials, OLS with Step Functions, and OLS. Using Cross Validation, I found the prediction method with the lowest MSE and used that prediction method to make predictions on home values based on different demographics. However, many other studies in the same field have concluded that not a single demographic but a demographic combined with other variables is the main indicator for home value. In this study, I will use the Ordinary Least Square model with polynomials to predict home value as it has the lowest Test MSE. My prediction overall, seems to be accurate and aligns well with real-life home values.

1. Introduction

For this project, I will be investigating my question: How can I predict Home Value based on different demographics? Essentially, I will be testing how different demographics play a role in determining the value of a home within the United States. My question is incredibly important as it points to possible discrimination within the United States housing market, as it could show how certain demographics like race may be associated with higher or lower home values. It could also be used for real estate developers, and real estate agents to estimate home values based on the demographic that currently resides inside the house. Overall my question is incredibly important as it has many legal and financial implications both for the government and for the real estate industry. In general, predicting price levels is incredibly important, so being able to predict price levels for the real estate market would be greatly beneficial for this industry.

The paper *Analyzing the Relationships among Race, Income, and Home Ownership in America* investigates the disparities in homeownership between white and minority households in the United States. The key findings from this study show no significant racial difference in homeownership rates among households who can meet mortgage down payment requirements. Overall, the study showed that white families were more likely to buy houses as they had larger wealth cushions that made it easier to buy homes even when incomes were similar. In comparison to my question this research focuses on home ownership, which has a relation to my topic. My question focuses more on actual home values, which may lead to a more thorough understanding of the real estate market and the role demographics play in it. My study will also involve more than just race, which will allow for a better understanding of different demographic variables and their effect on home values. I am also predicting home values, while they are not.

This study may also help explain my results in my study because they concluded that it is both wealth and race that affect home ownership, so that may in turn lead to the idea that a specific combination of demographics allows for accurate predictions of home values.

Another study, *Effects of Race and of Other Demographic Factors on the Values of Single-Family Homes* by Martin J. Bailey. Bailey used three demographic variables in his study: racial composition, population density, and income levels. Overall, he found that minority ownership did not lower values themselves, but it is associated with declining prices in the general area. This is a similar result to the research above. This relates to my question as well because it finds the relationship between race and values of single-family houses, however like the study above it mainly only focuses on a singular demographic. Additionally, this study helps to confirm that there may be a specific combination of demographics that may help me predict home values.

The paper *AGE, housing demand, and real house prices* by Richard Green, investigates the role of age in housing demand and housing prices. When Green was isolated from other factors like income and education, he could not find a significant decline in housing demand as people got older. He found that income and education levels to be the strongest predictors of housing demand. This study is in line with all studies above showing that a single demographic is not the reason for a significant impact on housing demand or value. In relation to my study, this study points to the idea that a predictor like total income is incredibly important, and will help predict home values.

The paper *Demographic Changes and the Housing Market* by Yifan Gong examines the extent to which demographic factors—rising life expectancy, international immigration, urbanization, and fertility rates—contribute to house price changes. From this study he found

these demographics accounted for 4.42%-18.8% growth in house price growth. Similarly to my question this paper examined age and the effect on housing price changes. He found as the life expectancy rises house prices tend to increase. This tells us that age can be a strong predictor of house value.

The paper *The Effect of Race on Property Values: The Case of Dallas* Overall, examines if race has a significant effect on property values in middle-income neighborhoods in Dallas, Texas. The study examines a transitioning black neighborhood and a white control neighborhood. The study finds that homes sold from white sellers to black buyers had higher prices. A problem with this study is the scale, in that they only tested in Dallas, Texas, and only tested two neighborhoods. My study will be examining people all over the United States, leading to more accurate results for predicting home values within the United States. Additionally, in relation to my question, this study shows that income and race may be predictive of house value. I see that because the study uses the predictors of race and middle-income neighborhoods.

1.1 Model

In my investigation I will be using R and using the model: $Y(\text{Home Value}) = \beta_0 + \beta_1 X_1(\text{Race}) + \beta_2 X_2(\text{Sex}) + \beta_3 X_3(\text{Family Size}) + \beta_4 X_4(\text{Total Family Income}) + \beta_5 X_5(\text{Age}) + \beta_6 X_6(\text{Years in the United States}) + \beta_7 X_7(\text{Employment Status}) + \beta_8 X_8(\text{Total Household Income}) + \beta_9 X_9(\text{Age}) + \beta_{10} X_{10}(\text{City}) + \varepsilon$. My data set is from IPUM, which combines data from the US Census and the American Community Survey(ACS). Through my investigation, I will be testing different models such as: OLS, OLS with Polynomials, OLS with a step function, LASSO, and Regression Trees.

2. Data

I obtained my data from IPUMS USA. IPUMS takes census data and survey data from around the world. IPNUM USA receives its data from the U.S. Census and the American Community Survey(ACS). This data is commonly used for demographic, social, and economic analysis. I created a custom data set on the IPNUMs website for 2021 involving, Race, Sex, Family Size, Total Family Income, City, Number of Own Family Members in Household, Age, Years in the United States, Employment Status, and Usual Hours Worked per Week. After removing the default columns the IPUMS that adds which were: Household Strata for Variance estimate, Group Quater Status, Personal Number in Sample Unit, and Person Weight the Sample Size came out to be 249,456. Before removing those columns the sample size was 332,608.

2.1 Summary Statistics

My data set has 20788 rows and 16 columns. Each row represents either an individual family unit or a person within the United States, so that is 20788 households. The variables *City*, *Sex*, *Race*, and *Employment Status* are categorical variables. However, they are already encoded with dummy variables provided by IPUM. For these categorical variables, Table 1 gives us the total for each category and not any other significant information. From Table 1, I can see that *Total Household Income*, *House Value*, *Usual Hours Worked per Week*, and *Total Family Income* have the largest spreads in the range. I can also see large differences in the mean and median, in that the mean is much greater than the median, pointing to a large right skew in these four variables. In age, *Family Size*, *Age*, and *Years In the United States* the mean and median appear much closer, which means the data here is much more normal and has little skew. The year variable is just for 2021, so it can be ignored.

3. Results

For this section of my report, I will focus on my results. I will use several prediction methods such as OLS, Logistic, Polynomials, Step Functions, Lasso, PCA, and Regression Trees. For each of the prediction methods, I used cross-validation to find the model within each method that results in the lowest Test MSE. Once that was completed, I compared the lowest Test MSE values from each method to find which method resulted in the overall lowest Test MSE.

3.1 OLS and Logistic

The first prediction method I tested was the Ordinary Least Square Method. Using this method with the combination of k-fold cross-validation where $k = 5$. I obtained a Test MSE of $1.560159e+13$ which can be seen in Table 3. The second prediction I tested was Logistic Regression, and with the same k-fold cross-validation, I obtained a significantly higher Test MSE. Through this calculation I decided to use OLS as it gave a lower Test MSE in comparison to Logistic regression.

3.2 Polynomial with OLS

When doing the polynomial prediction method on my model I obtained the result of the 5th degree having the lowest Test MSE. For the polynomial prediction method, I used cross-validation to find the MSE associated with each degree value for my model. I then found which model has the lowest Test MSE out of the five. Essentially, I created five models with degrees up to five, and then used cross-validation on each to find a Test MSE, and then picked the polynomial model with the lowest Test MSE. Using the polynomial model with the lowest Test MSE I obtained a Test MSE of $1.36e+13$.

3.3 Step Function with OLS

For the prediction method of step functions, I am taking my predictors and varying the number of cuts on each. I then used cross-validation for each number of cuts to obtain the Test MSE for each number of cuts. I used cuts from 2-10 and obtained that the 9th cut resulted in the lowest Test MSE of $1.527827e+13$.

3.4 LASSO

The next prediction method I conducted was LASSO. Using cross-validation I trained the model to go through my model with different values of lambda. The cross-validation portion split the model into many folds, trained the model on the training set, and then tested it on the validation set. After repeating this process I obtain the lambda value 10801.29 as the value that best minimizes Test MSE. This lambda value signals that there may be variable selection, and having many predictors set to 0. After conducting this part I fit my lambda value and calculated a Test MSE of $1.491816e+13$.

3.5 Regression Trees

The prediction method for regression trees shown in Figure 2 shows us that as it reaches a tree size of 6 it minimizes the cross-validation error the most. Figure 3 shows us that not all predictors are useful and that the predictors with the most predictive power are *Family Size*, *Age*, *Total Family income*, and *total household income*.

To find the best regression tree model, I created trees with different numbers of nodes. I then used cross-validation to find the number of nodes that minimized the cross-validation error the most. Since the size of 6 was the best tree, I then pruned the tree to six nodes. After pruning, I obtained a Test MSE of $1.458903e+13$.

3.6 PCA

From Figure 4, I can see that Principle Component 1(PC1) and PC2 have the most weight in the dataset, roughly covering 40% of the variance. PC4 through PC7 appears to have little to no explanation of the variance.

3.4 Best Prediction Model

The best model can be seen in Table 3, and the one with the lowest Test MSE is Polynomials with OLS.

Prediction

From Table 3, I am able to see that the prediction method using polynomials with OLS results in the lowest Test MSE. Using this predictive model I can predict home value. Using Table 5, I obtained a predicted home value of \$2,163,174. This price seems accurate for a home in San Jose, CA that is able to fit a family of four. According to SF Gate, the median home price in San Jose is \$2,008,000, which closely aligns with my prediction. The other input variables—such as a 40-year-old individual with a total family income of \$200,000, who is employed and works 40 hours per week—are reasonable factors in predicting this value, as someone with this background would likely own a home of similar worth. However, one problem with using the polynomial method is that I was unable to apply the polynomial with 5 as the value for *Employment Status* and *Sex* since they only had 2 and 4 options respectively. I am also unsure how much of a factor race and sex are for more prediction. I also found that when I change the city for my model the home value changes drastically, which may show a correlation between demographics and city and its use in prediction for home value.

Conclusion

Overall, through examination of my question: How to predict Home Values based on demographics, using data from IPUM, I compared six predictive models using out-of-sample data in order to find the model that would give the best accuracy for predicting Home values. Using cross-validation on each prediction method, allowed me to obtain Test MSEs for each model within each prediction method. This eventually allowed me to find which of the prediction models had the lowest Test MSE.

I found that the polynomial model using OLS and 5 dimensions had the lowest Test MSE. Using my model, I was able to successfully predict housing prices in San Francisco for a given person. Through my research I have discovered that it is possible to predict home values based on demographics, however, it is important to use multiple variables, as the city drastically changes the home price in my model.

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Appendix

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Year      City      Total_household_income  House_value  Family_Size  Sex      Age      Race      Years_in_the_United_States  Employment_status
Min. :2021  0      :18363  Min. : -5150  Min. : 1000  Min. : 1.00  1:10207  Min. : 0.00  1      :13480  Min. : 0.000  0:3408
1st Qu.:2021 4610 : 475  1st Qu.: 46800  1st Qu.: 200000  1st Qu.: 2.00  2:10581  1st Qu.:22.00  2      : 2133  1st Qu.: 0.000  1:9474
Median :2021 3730 : 252  Median : 88800  Median : 450000  Median : 3.00  Median :43.00  8      : 2127  Median : 0.000  2: 583
Mean :2021 1190 : 134  Mean : 602803  Mean :3239718  Mean : 2.99  Mean :42.78  7      : 1283  Mean : 3.534  3:7323
3rd Qu.:2021 5350 : 98  3rd Qu.: 157000  3rd Qu.:9999999  3rd Qu.: 4.00  3rd Qu.:63.00  6      : 960  3rd Qu.: 0.000
Max. :2021 2990 : 51  Max. :9999999  Max. :9999999  Max. :11.00  Max. :95.00  4      : 335  Max. :87.000
(Other): 1415
Usual_hours_worked_per_week  Total_family_income
Min. : 0.00  Min. : -5150
1st Qu.: 0.00  1st Qu.: 44000
Median : 2.00  Median : 85000
Mean :19.01  Mean : 615935
3rd Qu.:40.00  3rd Qu.: 154620
Max. :99.00  Max. :9999999

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Table 1: Summary statistics for all variables within my dataset

Variables	Description
<i>Race</i>	Race of a person
<i>Age</i>	Age of a person
<i>Sex</i>	Sex of a person
<i>Family Size</i>	Number of family members in the household
<i>Usual Hours Worked per Week</i>	Usual amount of hours worked per work for a person
<i>Total Family Income</i>	Total pre-tax income earned by one's family
<i>Years in the United States</i>	Total years a person has lived in the United States
<i>City</i>	City of person
<i>Employment Status</i>	Indication of employment status
<i>Year</i>	Year of Study
<i>Total Household Income</i>	Total income of all household members age 15+
<i>Home Value</i>	Value of housing unit

Table 2: Variables in the study

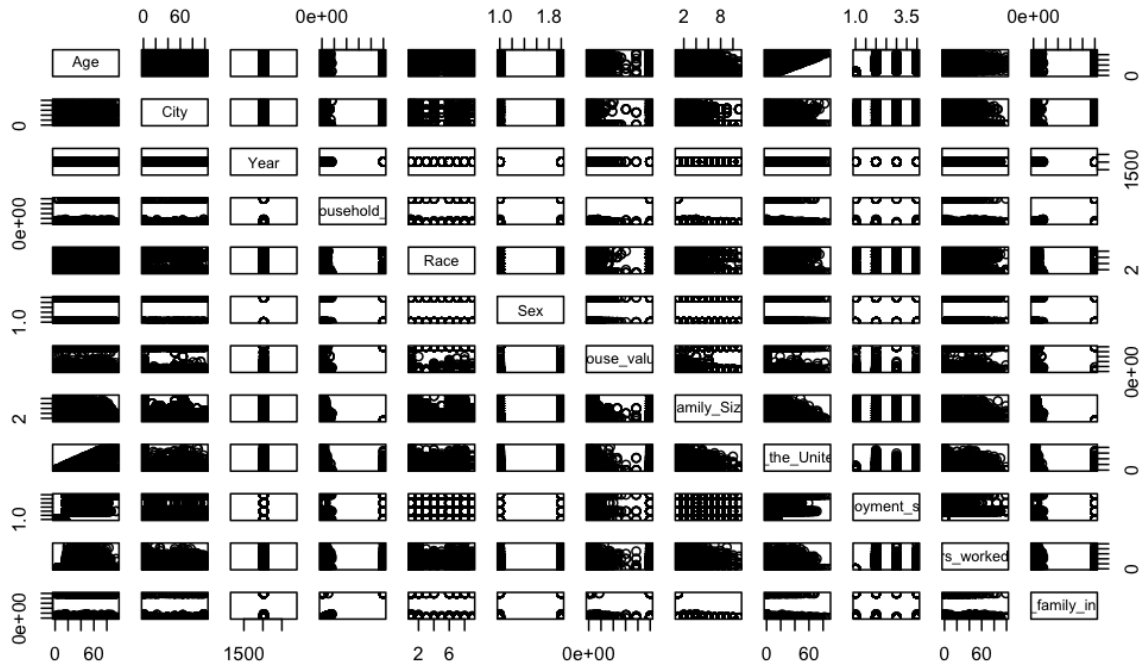


Figure 1: Scatter plot of raw data

Prediction Method	Test MSE
OLS	1.480526e+13
GLM	2.985188e+13
LASSO	1.491816e+13
Polynomial to the 5th degree	1.365497e+13
Step Function	1.527827e+13
Regression Tree	1.458903e+13

Table 3: Table of Test MSE values for each prediction method

d	Test MSE
1	1.566599e+13
2	1.462039e+13
3	1.411425e+13
4	1.378688e+13
5	1.365497e+13

Table 4: Polynomial Test MSE for the five degrees

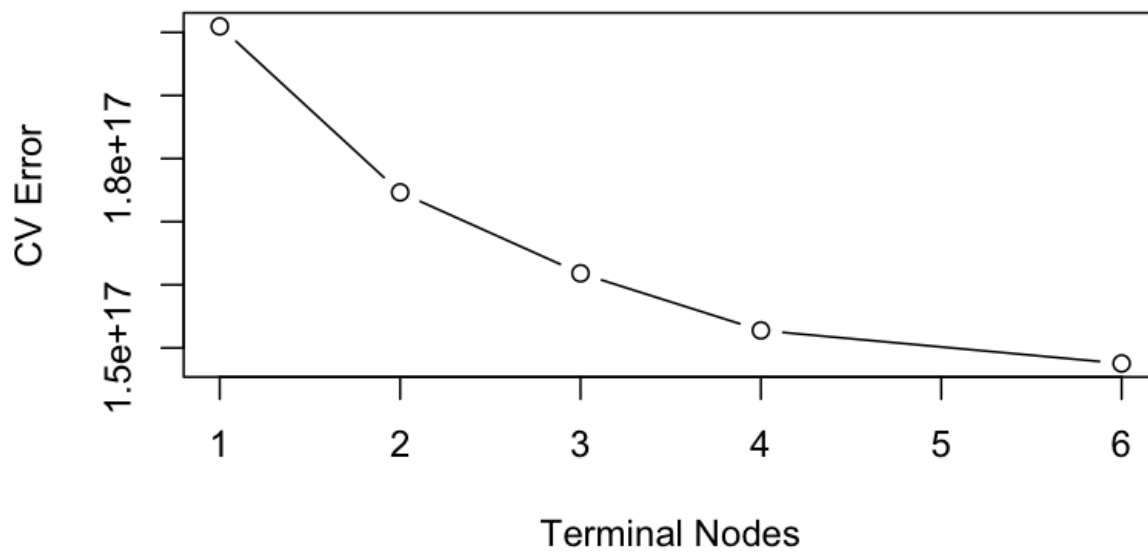


Figure 2: Cross-validation deviation compared to size

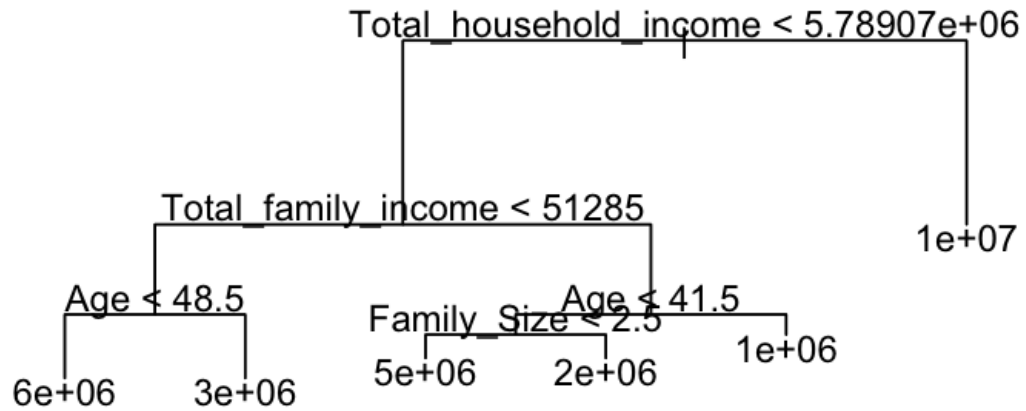


Figure 3: Regression tree containing the subset with the most predictive power

Variables	Prediction Value
Race	1 (White)
Age	40
Sex	1 (Female)
Family_Size	4
Usual_hours_worked_per_week	40
Total_family_income	\$200,000
Years_in_the_United_States	40
City	6310 (San Jose)
Employment_Status	2 (Employed)
Total_household_income	\$200,000

Table 5: Prediction input numbers to calculate predicted house value

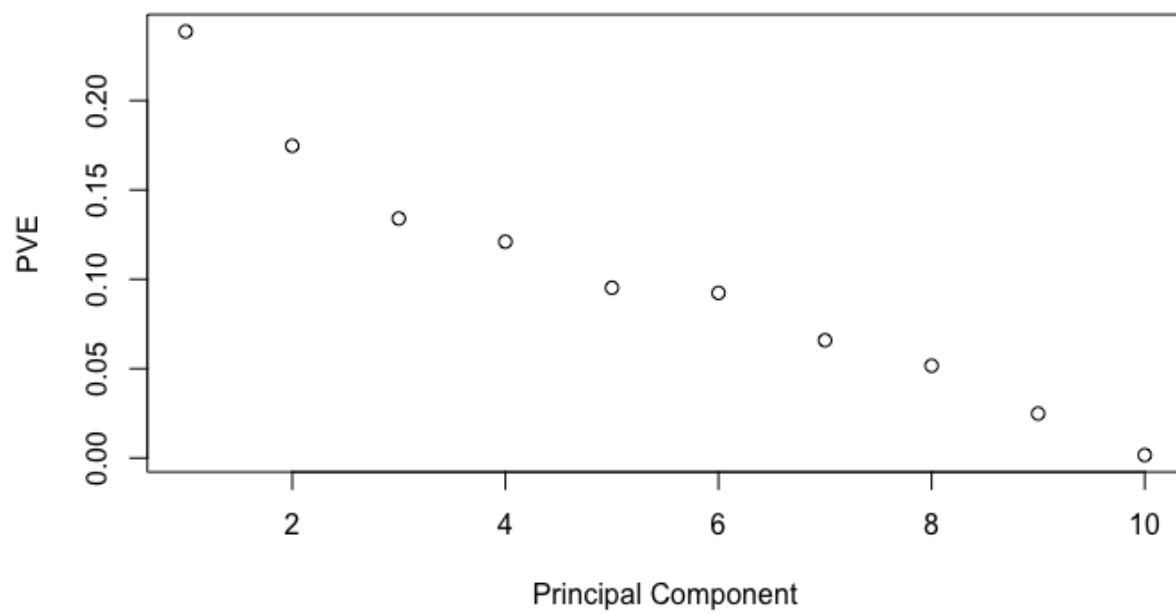


Figure 4: PCA plot